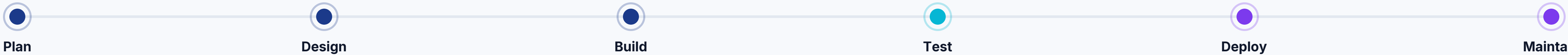


TECH STACK

- Python 3.12
- FastAPI
- Airflow 2.10
- SQLAlchemy
- NumPy
- Alembic
- WeasyPrint
- React 19
- TypeScript
- Tailwind
- Recharts
- PostgreSQL 16
- MySQL 8.4
- Argon2id + TOTP
- Server-Sent Events
- Docker Compose
- GitHub Actions
- pytest + mypy + Ruff

DEVELOPMENT TIMELINE



PROJECT OUTCOMES

Working pipeline	28-table warehouse, 20 views	12-rule DQ gate	REST API, 173 operations	408 automated tests
104 API smoke checks	React dashboard, 20 screens	Alembic-managed schema	30 documents, 80 diagrams	PDF and CSV reports